THE CUMULATIVE ATTRITION RISK HYPOTHESIS

STRAIN, EMBEDDEDNESS AND THE LIMITS OF PAY

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Introduction

Employee attrition is a major organizational problem because replacing workers requires recruitment, training, and time until new employees become productive. Yet employee turnover is difficult to explain, since leaving a job is rarely driven by one factor alone. Work pressure, compensation, seniority, and personal circumstances may all shape an employee's decision to stay or leave. Understanding these patterns is the central concern of HR analytics, the field that uses employee data to support workforce decisions.

Prior turnover research shows that pay is only one part of a broader retention problem (Griffeth, Hom, & Gaertner, 2000). Job embeddedness theory further suggests that employees with stronger personal, social, and organizational ties are less likely to leave, because leaving becomes more costly for them (Mitchell, 2001). More recent research on job demands and burnout also indicates that financial satisfaction may not fully offset the effect of high work pressure on turnover intentions (Wong, 2024).

Our research question is: **how is the accumulation of overtime, business travel, and single status associated with employee attrition, and does higher pay weaken this association?**

H1: Employee attrition increases as work-life strain markers (overtime, business travel and single status) accumulate - each additional marker is associated with higher odds of leaving.

H2: Higher pay is associated with lower attrition, but this association weakens as strain accumulates -pay does not fully offset the effect of high strain.

We treat overtime and business travel as direct work-life strain markers, while single status is used as a proxy for lower life embeddedness rather than as a strain factor by itself. We combine these two types of risk marker - work strain and low embeddedness - into a single index that we call strain load for brevity. Using the IBM dataset, we examine whether attrition rises as these markers accumulate.

Because Attrition is a binary outcome, we use logistic regression to estimate associations with the odds of leaving. Our contribution is to move beyond isolated risk factors by testing a cumulative-strain explanation and examining whether the apparent protective role of income remains under high strain. We interpret the findings as associations, not causal effects.

Data Overview

The dataset used in this project is the IBM HR Analytics Employee Attrition dataset, a synthetic dataset created by IBM data scientists for learning and research. The entity in our data is an employee: each row represents one employee in the company. The dataset includes 1,470 employees and 35 columns, with no missing values. We also verified the internal consistency of the data: there are no impossible age–experience combinations, confirming no data-quality issues in these fields. Our target variable is Attrition, indicating whether an employee left the company. The data is imbalanced: 237 employees left (16.1%), while 1,233 employees stayed (83.9%). We removed four non-informative columns that are either constant for all employees or only identifiers: EmployeeCount, Over18, StandardHours, and EmployeeNumber, leaving 31 variables: the target and 30 predictors.

Provided below is a list of the feature families used to tackle our research question:

Strain and embeddedness markers: These include **OverTime**, **BusinessTravel**, and **MaritalStatus**. OverTime and BusinessTravel represent pressure on employees' time and work-life balance, while single status, derived from MaritalStatus, is used as a proxy for lower life embeddedness.

Compensation feature: **MonthlyIncome** is used to examine whether higher pay is associated with lower attrition, especially when strain factors accumulate.

Seniority controls: Since income is often related to career stage, we use **Age**, **JobLevel** and **TotalWorkingYears** as a limited check for whether the income-attrition relationship is mainly explained by seniority.

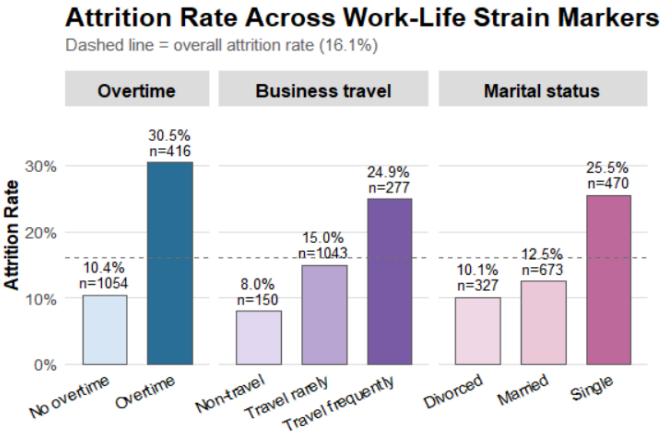
To support the analysis, we create two derived variables.

First - **StrainLoad** counts how many of the three risk markers apply to each employee: overtime, any business travel, and single status, producing a score from 0 to 3.

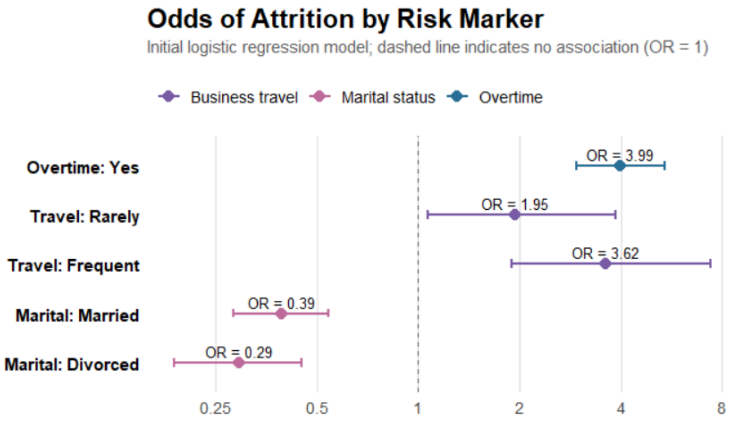
Second - MonthlyIncome is divided into three equal-sized pay groups (**low, medium, high**) so that attrition can be compared across pay levels within each strain level.

Methods and results

To answer the research question, we created two derived variables from the features described above: a StrainLoad score, counting overtime, any business travel, and single status from 0 to 3, and three equal-sized MonthlyIncome groups. Since Attrition is binary, we used logistic regression to estimate how each factor is associated with the odds of leaving. We report odds ratios with confidence intervals and interpret the results as associations rather than causal effects.



The initial logistic regression model included the three main risk markers: OverTime, BusinessTravel, and MaritalStatus. Overtime was the strongest marker: employees working overtime had about four times the odds of leaving compared with employees who did not work overtime. Business travel was also associated with higher attrition, especially frequent travel. Married and divorced employees had lower odds of leaving than single employees, consistent with the idea that single status may reflect lower life embeddedness. These results show that all three markers are relevant to attrition, but they do not yet show whether risk accumulates.

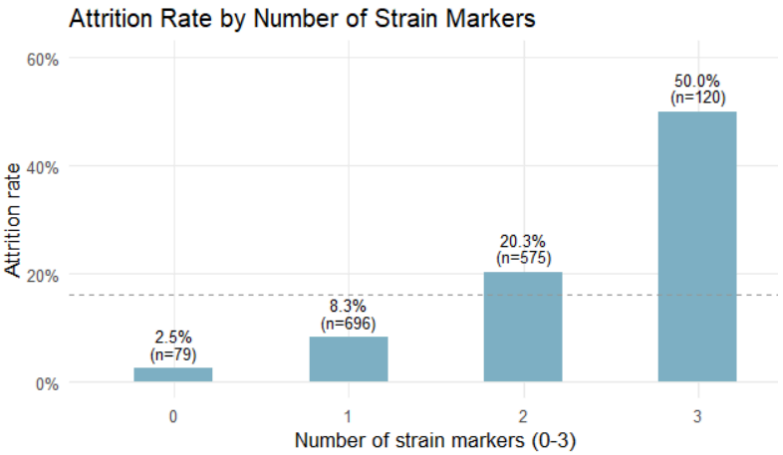


[link for logistic table]

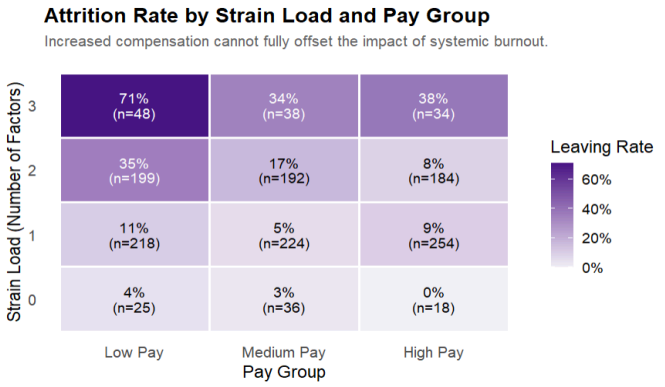
To test the cumulative-strain hypothesis, we used the StrainLoad score. Attrition rose sharply as more risk markers applied to the same employee: from 2.5% with no markers, to 8.3% with one, 20.3% with two, and 50.0% with all three. A logistic model using StrainLoad showed that each additional marker multiplied the odds of leaving by about 3.2, and the upward trend was statistically significant. We did not find evidence that giving the three markers separate weights improved model fit over the simple StrainLoad count (likelihood-ratio test, $p = 0.13$). This suggests that the simple accumulation score captures most of the observed attrition pattern.

[link for Cumulative Strain and Attrition Tests]

This monotonic rise supports our first hypothesis: attrition increases as strain markers accumulate.



We then examined whether the strain-attrition pattern differs by pay level. MonthlyIncome was associated with lower attrition when examined alone, but this association became weaker and no longer significant after adding Age, JobLevel, and TotalWorkingYears (seniority controls). This suggests that part of the income-attrition relationship reflects seniority rather than pay alone. Comparing attrition across pay groups within each StrainLoad level showed that higher pay was associated with lower attrition mainly when strain was low or moderate. At the highest strain level, attrition remained high even among medium- and high-pay employees, and the difference between these two groups was small. Because the highest-strain pay cells are relatively small, the exact percentages should be interpreted cautiously, but the overall pattern supports the same conclusion: income may be linked to lower attrition up to a point, but it does not fully offset accumulated strain.



Income OR after seniority controls

Model	Income OR	95% CI	p-value
Income only	0.55	0.45–0.67	<0.001
Income + Age	0.64	0.52–0.80	<0.001
Income + seniority controls	0.95	0.57–1.58	0.843

This pattern supports our second hypothesis: pay is associated with lower attrition, but its protective association weakens under high strain.

As a descriptive model check, the three-marker logistic model reached an in-sample AUC of 0.73, indicating moderate separation between employees who stayed and employees who left. Since the goal of this project is explanation rather than individual prediction, we use the model mainly to understand the direction and strength of associations.

Limitations and Future Work

This analysis has several limitations. First, the IBM HR Analytics dataset is synthetic, so the observed patterns may partly reflect how the data were constructed rather than real workplace dynamics. Second, the data are cross-sectional, meaning that we can identify associations between strain markers, pay, and attrition, but cannot claim that strain causes employees to leave or that pay prevents leaving. Third, our strain measure is indirect: overtime and business travel are proxies for work-life pressure, while single status is used as a proxy for lower life embeddedness; combining these two mechanisms into one score is therefore a simplification. Fourth, the dataset does not distinguish between voluntary resignations and dismissals, which limits retention-related conclusions. Finally, after splitting the highest-strain group by pay level, some cells are relatively small, so the exact percentages should be interpreted cautiously. These issues limit how far the findings can be generalized beyond this dataset.

With more time and access to richer data, we would validate the same pattern on a real organizational dataset, include direct measures such as job satisfaction, burnout, workload history, and team-level factors, and use survival analysis to study not only whether employees leave, but also when they leave.

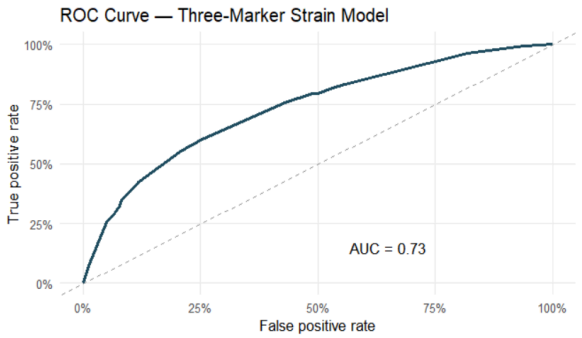
Appendix

Reproducibility Links **Code repository:** GitHub | **Data:** Kaggle - IBM HR

References

- Griffeth, Hom, & Gaertner (2000). A meta-analysis of employee turnover antecedents. Journal of Management, 26(3), 463–488.
- Mitchell et al. (2001). Job embeddedness and voluntary turnover. Academy of Management Journal, 44(6), 1102–1121.
- Wong et al. (2024). Job demands, burnout, pay satisfaction, and turnover intention. Journal of Nursing Management, 2024, 3534750.

Model Checks



Logistic regression odds ratios.

Reference groups: no overtime, non-travel, single.

Variable	OR	95% CI	p-value
Overtime: Yes	3.99	2.97–5.39	<0.001
Travel: Rarely	1.95	1.07–3.85	0.040
Travel: Frequent	3.62	1.90–7.42	<0.001
Marital: Married	0.39	0.28–0.54	<0.001
Marital: Divorced	0.29	0.19–0.45	<0.001

Statistic	Result
OR per additional strain marker	3.24 (95% CI 2.62-4.01), p < 0.001
Cochran-Armitage trend test	chi-square = 130.7, p < 0.001
LRT: separate weights vs. simple count	p = 0.13 (no improvement)